

| Question |     |      |  | Marking details                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) | (i)  |  | (each carbon atom) only bonded to 3 other carbon atoms (1)<br><br>do not award first mark if any reference to metallic bonding<br><br>delocalised electrons able to move (through structure) (1)                                                    | 2               |     |     | 2     |       |      |
|          |     | (ii) |  | $9.1 \times 10^{-10}$ (3)                      accept $0.91 \times 10^{-9}$<br><br>if incorrect award (1) for each of following<br><br>$11 \times 0.26 = 2.86$<br><br>diameter = circumference $\div \pi$ / $\frac{2.86}{3.14}$<br><br>ecf possible |                 | 3   |     | 3     | 3     |      |
|          |     |      |  | Question 4 total                                                                                                                                                                                                                                    | 4               | 5   | 0   | 9     | 3     | 0    |